



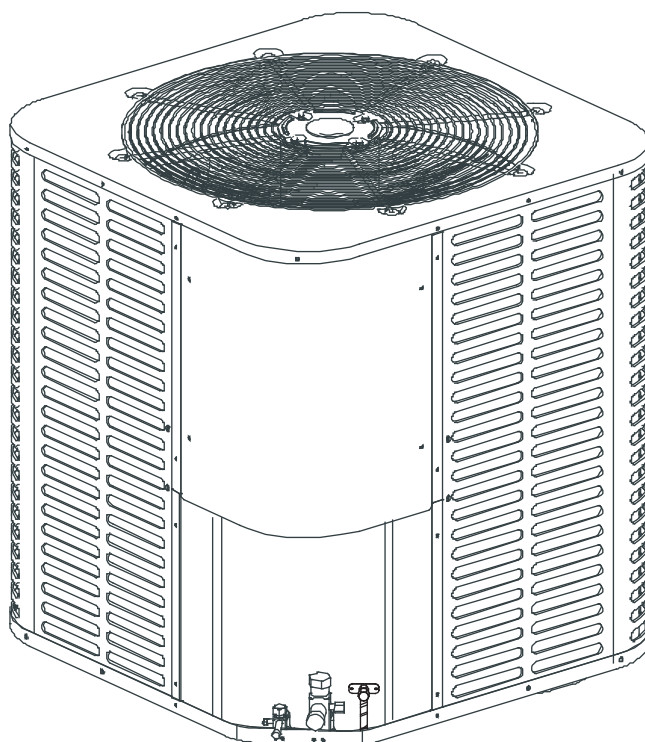
INSTALLATION INSTRUCTIONS

13 SEER

Split System Air Conditioner

1.5-5 Tons

R410A



NOTE: Appearance of unit may vary.



RECOGNIZE THIS SYMBOL AS AN INDICATION OF IMPORTANT SAFETY INFORMATION



WARNING

These instructions are intended as an aid to qualified licensed service personnel for proper installation, adjustment and operation of this unit. Read these instructions thoroughly before attempting installation or operation. Failure to follow these instructions may result in improper installation, adjustment, service or maintenance possibly resulting in fire, electrical shock, property damage, personal injury or death.



DO NOT DESTROY THIS MANUAL

Please read carefully and keep in a safe place for future reference by a serviceman.

TABLE OF CONTENTS

1.0 SAFETY.....	3
1.1 INSPECTION.....	4
1.2 LIMITATIONS.....	4
2.0 GENERAL.....	4
3.0 UNIT INSTALLATION.....	6
3.1 LOCATION.....	6
3.2 GROUND INSTALLATION.....	6
3.3 ROOF INSTALLATION.....	6
3.4 UNIT PLACEMENT.....	6
3.5 UNIT MOUNTING.....	7
3.6 FACTORY-PREFERRED TIE-DOWN METHOD.....	8
3.7 PRECAUTIONS DURING LINE INSTALLATION.....	9
3.8 PRECAUTIONS DURING BRAZING OF LINES.....	10
3.9 PRECAUTIONS DURING BRAZING SERVICE VALVE.....	10
4.0 INTERCONNECTING TUBING.....	12
4.1 SUCTION AND LIQUID LINES	12
4.2 MAXIMUM LENGTH OF LINES	12
4.3 VERTICAL LIFT	12
5.0 EVACUATION.....	12
6.0 ELECTRICAL CONNECTIONS.....	14
6.1 GENERAL INFORMATION & GROUNDING	14
6.2 FIELD CONNECTIONS POWER WIRING	14
6.3 REMOVING THE TOP PANEL AND MOTOR.....	15
7.0 CHECKING REFRIGERANT CHARGE	15
7.1 CHARGING BY LIQUID PRESSURE.....	15
7.2 CHARGING BY WEIGHT	16
7.3 FINAL LEAK TESTING	16
8.0 WARRANTY.....	16
9.0 WIRING DIAGRAM.....	16
9.1 CONTROL WIRING	17

This document is customer property and is to remain with this unit. These instructions do not cover all the different variations of systems nor does it provide for every possible contingency to be met in connection with installation.

All phases of this installation must comply with **NATIONAL, STATE, AND LOCAL CODES**. If additional information is required please contact your local distributor.

1.0 SAFETY



This is a safety alert symbol. When you see this symbol on labels or in manuals, be alert to the potential for personal injury.



This is an attention alert symbol. When you see this symbol on labels or in manuals, be alert to the potential for personal injury.

Understand and pay particular attention to the signal words **DANGER, WARNING, or CAUTION**.

DANGER indicates an imminently hazardous situation, which, if not avoided, will result in death or serious injury.

WARNING indicates a potentially hazardous situation, which, if not avoided, could result in death or serious injury.

CAUTION indicates a potentially hazardous situation, which, if not avoided may result in minor or moderate injury. It is also used to alert against unsafe practices and hazards involving only property damage.



WARNING

Improper installation may create a condition where the operation of the product could cause personal injury or property damage.
Improper installation, adjustment, alteration, service or maintenance can cause injury or property damage. Refer to this manual for assistance or for additional information, consult a qualified contractor, installer or service agency.



CAUTION

This product must be installed in strict compliance with the installation instructions and any applicable local, state, and national codes including, but not limited to building, electrical, and mechanical codes.



WARNING

FIRE OR ELECTRICAL HAZARD

Failure to follow the safety warnings exactly could result in serious injury, death or property damage.

A fire or electrical hazard may result causing property damage, personal injury or loss of life.



CAUTION

If using existing refrigerant lines make certain that all joints are brazed, not soldered.



CAUTION

Scroll compressor dome temperatures may be hot. Do not touch the top of compressor; it may cause minor to severe burning.

1.1 INSPECTION

As soon as a unit is received, it should be inspected for possible damage during transit. If damage is evident, the extent of the damage should be noted on the carrier's delivery receipt. A separate request for inspection by the carrier's agent should be made in writing. See Local distributor for more information.

Requirements For Installing/Serviceing R410A Equipment

- Gauge sets, hoses, refrigerant containers, and recovery system must be designed to handle the POE or PVE type oils.
- Manifold sets should be 800 PSIG high side and 250 PSIG low side with 550 PSIG low side restart.
- All hoses must have a 700 PSIG service pressure rating.
- Leak detectors should be designed to detect R410A.
- Recovery equipment (including refrigerant recovery containers) must be specifically designed to handle R410A.
- **Do not use an R-22 TXV.**
- Good Refrigeration practices require the installation of a field supplied liquid line drier, as shown in Fig.1.

NOTE :

It will be more convenient to open the Service valve after removing the Underside Clapboard.

Vapor Line Service Valve

Liquid Line Service Valve

LIQUID-LINE FILTER-DRIER
FIELD SUPPLIED AND INSTALLED

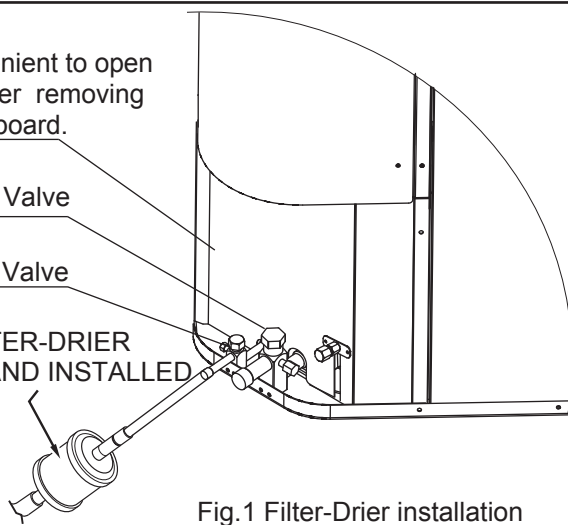


Fig.1 Filter-Drier installation

1.2 LIMITATIONS

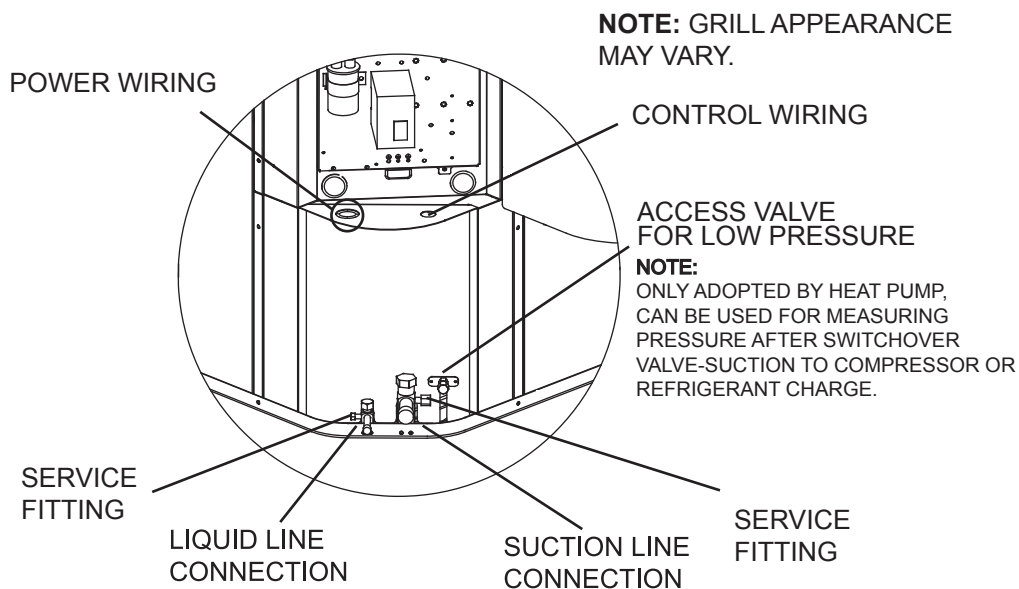
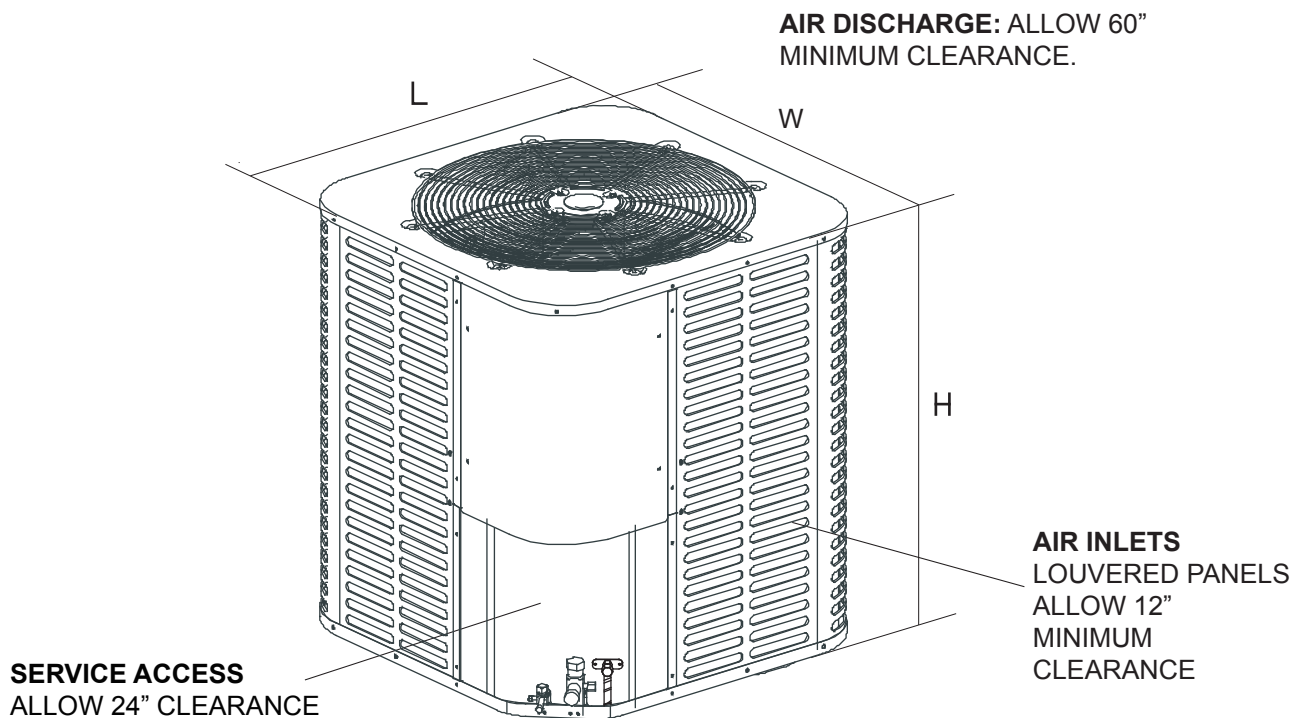
The unit should be installed in accordance with all National, State, and Local Safety Codes and the limitations listed below:

- 1.Limitations for the indoor unit, coil and appropriate accessories must also be observed.
- 2.The outdoor unit must not be installed with any duct work in the air stream. The outdoor fan is the propeller type and is not designed to operate against any additional external static pressure.
- 3.The maximum and minimum conditions for operation must be observed to assure a system that will give maximum performance with minimum service.
- 4.This unit is not designed to operate with a low ambient kit. Do not modify the control system to operate with any kind of low ambient kit.
- 5.The maximum allowable line length for this product is 150 feet.

2.0 GENERAL

The outdoor units are designed to be connected to a matching indoor coil with braze connect lines. Units are factory charged with refrigerant for a matching indoor coil plus 15 feet of field supplied lines.

Matching indoor coils are available with a thermostatic expansion valve or an orifice for the most common usage. The orifice size and/or refrigerant charge may need to be changed for some indoor-outdoor unit combinations, elevation differences or total line lengths.



DIMENSIONAL DATA

FIG.2 DIMENSIONS

MODEL SIZE	Dimensions (Inches)			Refrigerant Connection Service Valve Size	
	"H" in. [mm]	"W" in. [mm]	"L" in. [mm]	Liquid in.	Suction in.
18/24	24-15/16[633]	21-7/8[554]	21-7/8[554]	3/8	3/4
30	24-15/16[633]	29-1/8[740]	29-1/8[740]	3/8	3/4
36	24-15/16[633]	23-5/8[600]	23-5/8[600]	3/8	3/4
42	33-3/16[843]	29-1/8[740]	29-1/8[740]	3/8	3/4
43	33-3/16[843]	28[710]	28[710]	3/8	3/4
48	29-7/8[759]	28[710]	28[710]	3/8	7/8
60	33-3/16[843]	29-1/8[740]	29-1/8[740]	3/8	7/8

3.0 UNIT INSTALLATION

3.1 LOCATION

Before starting the installation, select and check the suitability of the location for both the indoor and outdoor unit. Observe all limitations and clearance requirements. The outdoor unit must have sufficient clearance for air entrance to the condenser coil, for air discharge and for service access. See Fig.2



NOTE

For multiple unit installations, units must be spaced a minimum of 24 inches apart. (Coil face to coil face.)

If the unit is to be installed on a hot sun exposed roof or a black-topped ground area, the unit should be raised sufficiently above the roof or ground to avoid taking the accumulated layer of hot air into the outdoor unit.

Provide an adequate structural support.

3.2 GROUND INSTALLATION

The unit may be installed at ground level on a solid base that will not shift or settle, causing strain on the refrigerant lines and possible leaks. Maintain the clearances shown in Fig.2 and install the unit in a level position.

Normal operating sound levels may be objectionable if the unit is placed directly under windows of certain rooms (bedrooms, study, etc.).

Top of unit discharge area must be unrestricted for at least 60 inches above the unit.



WARNING

The outdoor unit should not be installed in an area where mud or ice could cause personal injury or system damage.

Elevate the unit sufficiently to prevent any blockage of the air entrances by snow in areas where there will be snow accumulation. Check the local weather bureau for the expected snow accumulation in your area. Isolate the unit from rain gutters to avoid any possible wash out of the foundation.

3.3 ROOF INSTALLATION

When installing units on a roof, the structure must be capable of supporting the total weight of the unit, including a padded frame unit, rails, etc., which should be used to minimize the transmission of sound or vibration into the conditioned space.

3.4 UNIT PLACEMENT

1. Provide a base in the pre-determined location.
2. Remove the shipping carton and inspect for possible damage.
3. Compressor tie-down nuts should remain tightened.
4. Position the unit on the base provided.



CAUTION

This system uses R410A refrigerant which operates at higher pressure than R-22. No other refrigerant may be used in this system. Gauge sets, hoses, refrigerant containers, and recovery system must be designed to handle R410A. If you are unsure, consult the equipment manufacturer.

The outdoor unit must be connected to the indoor coil using field supplied refrigerant grade copper tubing that is internally clean and dry. Units should be installed only with the tubing sizes for approved system combinations. The refrigerant charge shown in the nameplate is for standard size interconnecting liquid line lengths up to 15 feet.



NOTE

Using a larger than specified line size could result in oil return problems. Using a too small line will result in loss of capacity and other problems caused by insufficient refrigerant flow. Slope horizontal suction lines at least 1" every 20 feet toward the outdoor unit to facilitate proper oil return.

3.5 UNIT MOUNTING

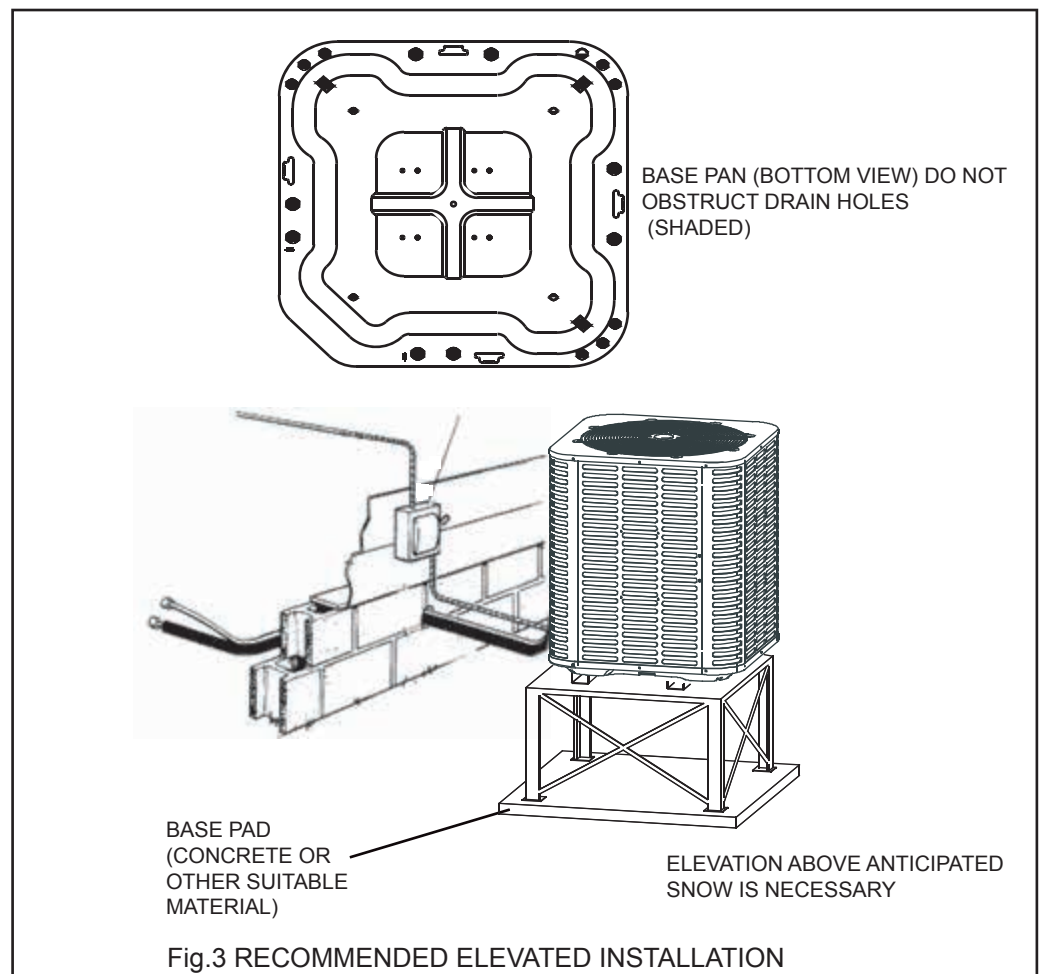
If elevating the heat pump, either on a flat roof or on a slab, observe the following guidelines.

1. The base pan provided elevates the heat pump 2" above the base pad.
2. If elevating a unit on a flat roof, use 4"× 4" (or equivalent) stringers positioned to distribute unit weight evenly and prevent noise and vibration (See Fig.3).

NOTE: Do not block drain openings shown in Fig.3.

3. If unit must be elevated because of anticipated snow fall, secure unit and elevating stand such that unit and/or stand will not tip over or fall off.

NOTE: To tie down unit, see 3.6.



3.6 FACTORY-PREFERRED TIE-DOWN METHOD

Step 1: Prior to installing clear pad of debris.

IMPORTANT

The pad must meet local codes and must be the proper thickness to accommodate fasteners.

Step 2: Center and level unit onto pad.

Step 3: Using field supplied L-shaped bracket to locate holes on concrete and drill pilot holes which is at least 1/4" deeper than fastener being used.

IMPORTANT

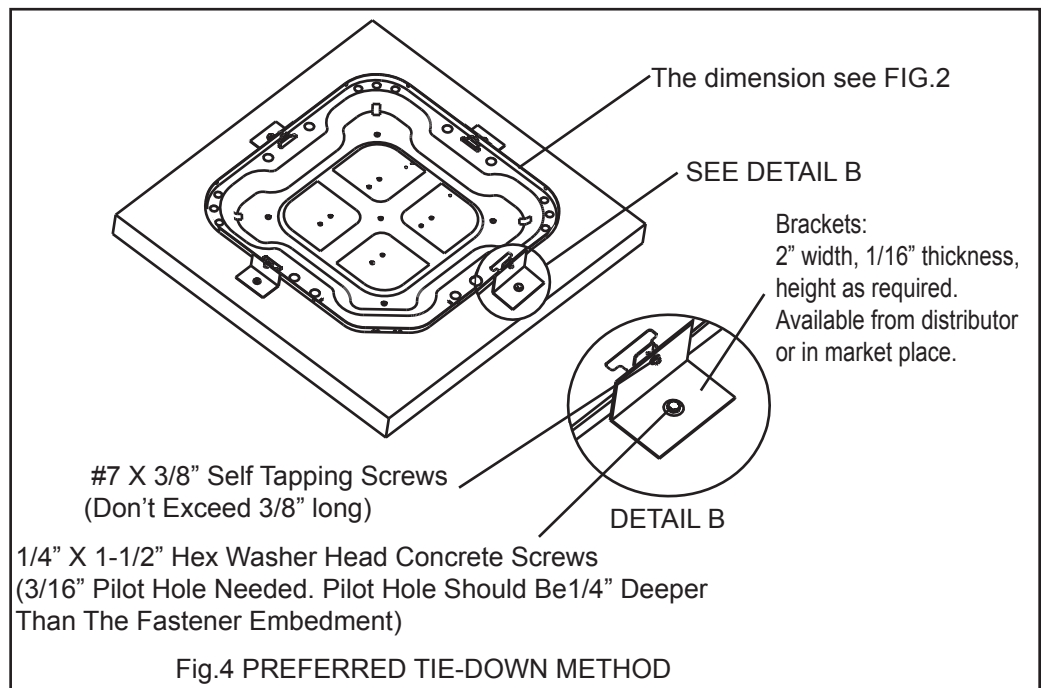
Self drilling screws to base pan should not exceed 3/8" long to avoid damaging coil.

Step 4: Using conventional practices to install brackets, tighten concrete fasteners and self-tapping screws (See Fig.4).

NOTE: 1. One bracket for each side. For extra stability, 2 brackets for each side.
2. Do not over-tighten the concrete fastener to avoid weakening the concrete.

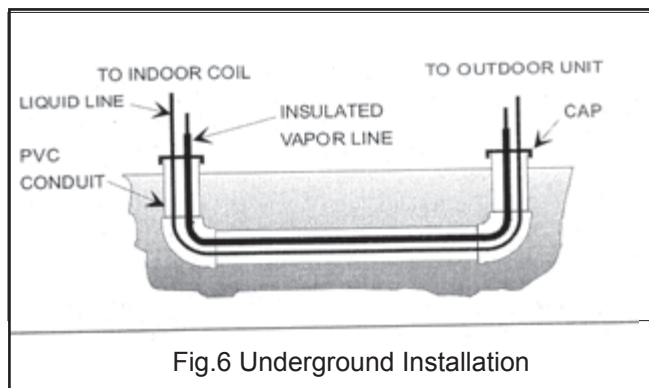
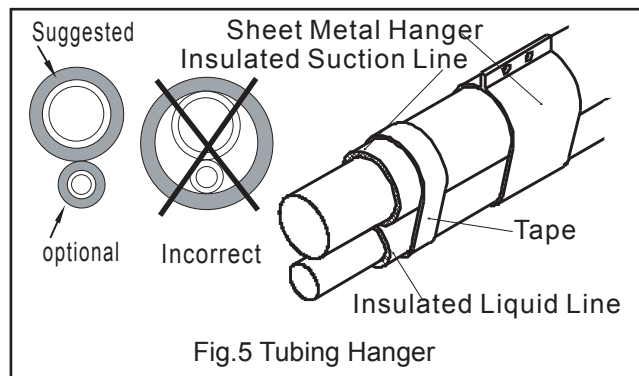
IMPORTANT NOTE:

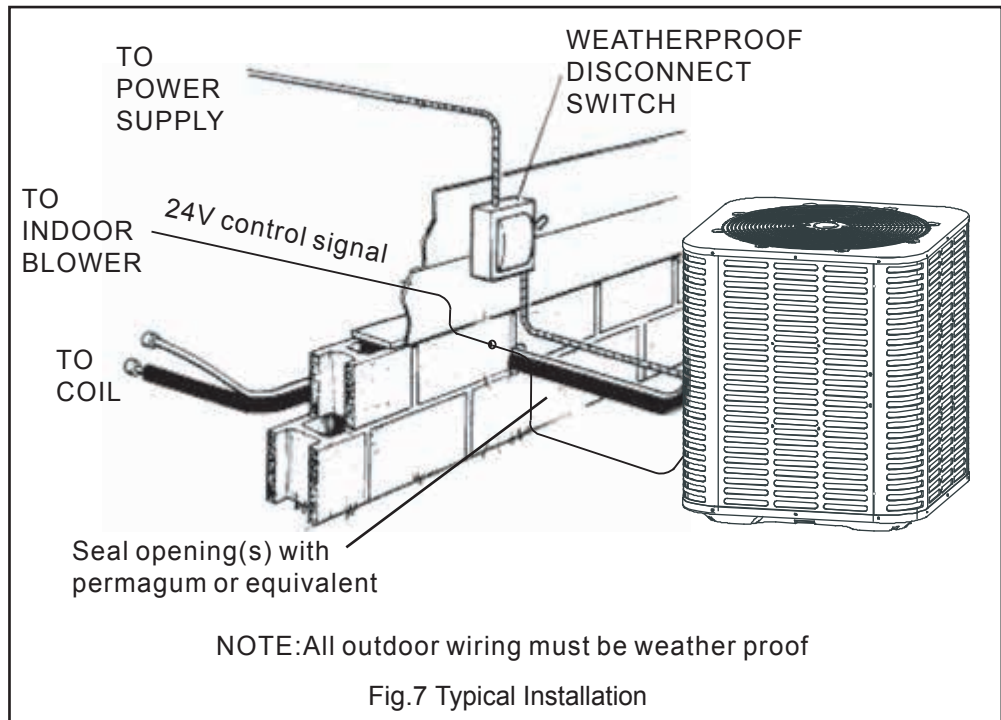
These instructions are intended to provide a method to tie-down system to cement slab as a securing procedure for high wind areas. It is recommended to check Local Codes for tie-down methods and protocols.



3.7 PRECAUTIONS DURING LINE INSTALLATION

1. Install the lines with as few bends as possible. Care must be taken not to damage the couplings or kink the tubing. Use clean hard drawn copper tubing where no appreciable amount of bending around obstruction is necessary, if soft copper must be used, care must be taken to avoid sharp bends which may cause a restriction.
2. The lines should be installed so that they will not obstruct service access to the coil, air handling system or filter.
3. Care must also be taken to isolate the refrigerant lines to minimize noise transmission from the equipment to the structure.
4. The suction line must be insulated. Tape and suspend the refrigerant lines as shown. DO NOT allow tube metal-to-metal contact. See Fig.5.
5. Use PVC piping as a conduit for all underground installations as shown in Fig.6. Buried lines should be kept as short as possible to minimize the build up of liquid refrigerant in the suction line during long periods of shutdown.
6. Pack a sealing material such as perma gum around refrigerant lines where they penetrate a wall to reduce vibration and to retain some flexibility.





3.8 PRECAUTIONS DURING BRAZING OF LINES

All outdoor unit and evaporator coil connections are copper-to-copper and should be brazed with a phosphorous-copper alloy material such as Silfos-5 or equivalent. DO NOT use soft solder. The outdoor units have reusable service valves on both the liquid and suction connections. The total system refrigerant charge is retained within the outdoor unit during shipping and installation. The reusable service valves are provided to evacuate and charge per this instruction.

Serious service problems can be avoided by taking adequate precautions to assure an internally clean and dry system.

	CAUTION
Dry nitrogen should always be supplied through the tubing while it is being brazed, because the temperature required is high enough to cause oxidation of the copper unless an inert atmosphere is provide. The flow of dry nitrogen should continue until the joint has cooled. Always use a pressure regulator and safety valve to insure that only low pressure dry nitrogen is introduced into the tubing. Only a small flow is necessary to displace air and prevent oxidation.	

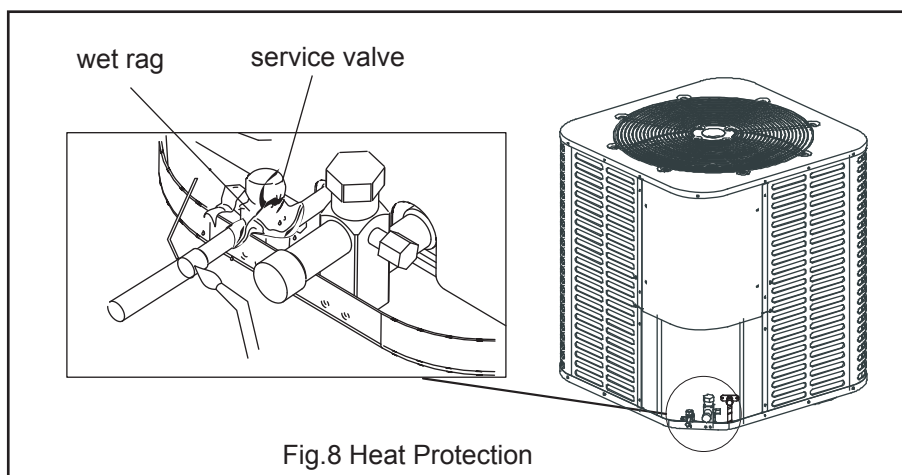
3.9 PRECAUTIONS DURING BRAZING SERVICE VALVE

Precautions should be taken to prevent heat damage to service valve by wrapping a wet rag around it as shown in Fig.8. Also, protect all painted surfaces, insulation, during brazing. After brazing cool joint with wet rag.

The valve can be opened by removing the plunger cap and fully inserting a hex wrench into the stem and backing out counter-clockwise until valve stem just touches the chamfered retaining wall.

Connect the refrigerant lines using the following procedure:

1. Remove the cap and Schrader core from both the liquid and suction service valve service ports at the outdoor unit. Connect low pressure nitrogen to the liquid line service port.



2. Braze the liquid line to the liquid valve at the outdoor unit. Be sure to wrap the valve body with a wet rag. Allow the nitrogen to continue flowing. Refer to the Tabular Data Sheet for proper liquid line sizing.
3. Carefully remove the rubber plugs from the evaporator liquid and suction connections at the indoor coil.
4. Braze the liquid line to the evaporator liquid connection. Nitrogen should be flowing through the evaporator coil.
5. Slide the plastic cap away from the suction connection at the indoor coil. Braze the suction line to the evaporator suction connection. Refer to the Table 1 for proper suction line sizing.
6. Protect the suction valve with a wet rag and braze the suction line connection to the outdoor unit. The nitrogen flow should be exiting the system from the suction service port connection. After this connection has cooled, remove the nitrogen source from the liquid fitting service port.
7. Replace the Schrader core in the liquid and suction valves.
8. Leak test all refrigerant piping connections including the service port flare caps to be sure they are leak tight. **DO NOT OVER TIGHTEN (between 40 and 60 inch -lbs. maximum).**
9. Evacuate the suction line, evaporator, and the liquid line, to 350 microns or less.

Table 1: Recommended Liquid and Suction Tube Diameters (In.)

MODEL SIZE	LIQUID	SUCTION
	Tube Diameter	Tube Diameter
18	3/8	3/4
24	3/8	3/4
30	3/8	3/4
36	3/8	3/4
42/43	3/8	3/4
48	3/8	7/8
60	3/8	7/8

10. Replace cap on service ports. Do not remove the flare caps from the service ports except when necessary for servicing the system.
11. Release the refrigerant charge into the system. Open both the liquid and suction valves by removing the plunger cap and with an hex wrench back out counter-clockwise until valve stem just touches the chamfered retaining wall.
12. Replace plunger cap finger tight, then tighten an additional 1/12 turn (1/2 hex flat). Cap must be replaced to prevent leaks.



WARNING

Never attempt to repair any brazed connections while the system is under pressure. Personal injury could result.

See "System Charge" section for checking and recording system charge.

4.0 INTERCONNECTING TUBING

4.1 SUCTION AND LIQUID LINES

Keep all lines sealed until connection is made.

Make connections at the indoor coil first.

Refer to Line Size Information in Tables 2 and 3 for correct size and multipliers to be used to determine capacity for various suction line diameters and lengths of run. The losses due to the lines being exposed to outdoor conditions are not included.

The factory refrigerant charge in the outdoor unit is sufficient for 15 feet of standard size interconnecting liquid line. Calculate actual charge required with installed liquid line size and length as below.

5/16" ± .4 oz. per foot

3/8" ± .6 oz. per foot

1/2" ± 1.2 oz. per foot

4.2 MAXIMUM LENGTH OF LINES

The maximum length of interconnecting line is 150 feet.

Always use the shortest length possible with a minimum number of bends.

NOTE: Excessively long refrigerant lines cause loss of equipment capacity.

4.3 VERTICAL LIFT

Keep the vertical lift to a minimum. Use the following guidelines when installing the unit:

1. DO NOT exceed the vertical lift as indicated on Table 3.
2. It is recommended to use the smallest liquid line size permitted to minimize system charge which will maximize compressor reliability.
3. Table 3 may be used for sizing horizontal runs.

5.0 EVACUATION

It will be necessary to evacuate the system to 350 microns or less. If a leak is suspected, leak test with dry nitrogen to locate the leak. Repair the leak and test again. To verify that the system has no leaks, simply close the valve to the vacuum pump suction to isolate the pump and hold the system under vacuum. Watch the micron gauge for a few minutes. If the micron gauge indicates a steady and continuous rise, it's an indication of a leak. If the gauge shows a rise, then levels off after a few minutes and remains fairly constant, it's an indication that the system is leak free but still contains moisture and may require further evacuation if the reading is above 350 microns.

LINE SIZING

TABLE 2: SUCTION LINE LENGTH/SIZE VS CAPACITY MULTIPLIER(R410A)

Model Size		1 1/2 Ton	2 Ton	2 1/2 Ton	3 Ton	3 1/2 Ton	4 Ton	5 Ton
Suction Line Connection Size		3/4" O.D.	3/4" O.D.	3/4" O.D.	3/4" O.D.	3/4" O.D.	7/8" O.D.	7/8" O.D.
Suction Line Run—Feet		5/8 Opt.	5/8 Opt.	5/8 Opt.	5/8 Opt.	5/8 Opt.	3/4 Opt.	1 1/8 Opt.
		3/4* Std.	3/4* Std.	3/4* Std.	3/4* Std.	3/4* Std.	7/8* Std.	7/8* Std.
25'	Optional	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	Standard	1.00	1.00	1.00	1.00	1.00	1.00	0.99
50'	Optional	0.97	0.97	0.97	0.97	0.98	0.98	0.99
	Standard	0.98	0.98	0.98	0.99	0.98	0.98	0.98
100'	Optional	0.94	0.94	0.94	0.94	0.95	0.95	0.98
	Standard	0.95	0.95	0.96	0.97	0.97	0.97	0.94
150'	Optional	0.90	0.90	0.90	0.90	0.91	0.92	0.97
	Standard	0.92	0.92	0.93	0.96	0.96	0.96	0.90

NOTES:

* Standard size

Using suction line larger than shown in chart will result in poor oil return and is not recommended.

the system capacity published in AHRI is 17800Btu/h. This data is based on the AHRI test condition and 25ft line set with standard suction tube size(3/4" tube). If you need to add the line set more than 25ft or use the optional suction tube size, you need to recalculate the capacity with the coefficient in the table. For example: for 50ft line and 5/8" suction tube, the capacity will be 17800*0.97=17266Btu/h.

TABLE 3 :LIQUID LINE SIZE (R410A)

Model Size	Line Size Connection Size (Inch O.D.)	Compressor Type	Line Size Connection And Line Size (Inch O.D.)	Liquid Line Size Outdoor unit above or below indoor coil					
				Total Equivalent Length - Feet					
				25	50	75	100	125	150
				Maximum Vertical Separation - Feet					
1 1/2 Ton	3/8"	Scroll	1/4	25	40	25	9	N/A	N/A
			5/16	25	50	60	58	40	30
			3/8*	25	50	60	60	40	30
2 Ton	3/8"	Scroll	1/4	23	N/A	N/A	N/A	N/A	N/A
			5/16	25	36	29	23	16	9
			3/8*	25	50	60	60	40	30
2 1/2 Ton	3/8"	Scroll	1/4	25	N/A	N/A	N/A	N/A	N/A
			5/16	25	49	38	27	17	6
			3/8*	25	50	60	60	40	30
3 Ton	3/8"	Scroll	5/16	25	50	37	22	7	N/A
			3/8*	25	50	60	60	40	30
3 1/2 Ton	3/8"	Scroll	5/16	25	23	4	N/A	N/A	N/A
			3/8*	25	50	43	36	30	24
4 Ton	3/8"	Scroll	3/8*	25	46	38	30	22	15
			1/2	25	50	56	55	40	30
5 Ton	3/8"	Scroll	3/8*	25	50	56	44	32	20
			1/2	25	50	60	60	40	30

NOTES:

* Standard line size

N/A Application not recommended.

6.0 ELECTRICAL CONNECTIONS

6.1 GENERAL INFORMATION & GROUNDING

Check the electrical supply to be sure that it meets the values specified on the unit nameplate and wiring label.

Power wiring, control (low voltage) wiring, disconnect switches and over current protection must be supplied by the installer. Wire size should be sized per requirements.



CAUTION

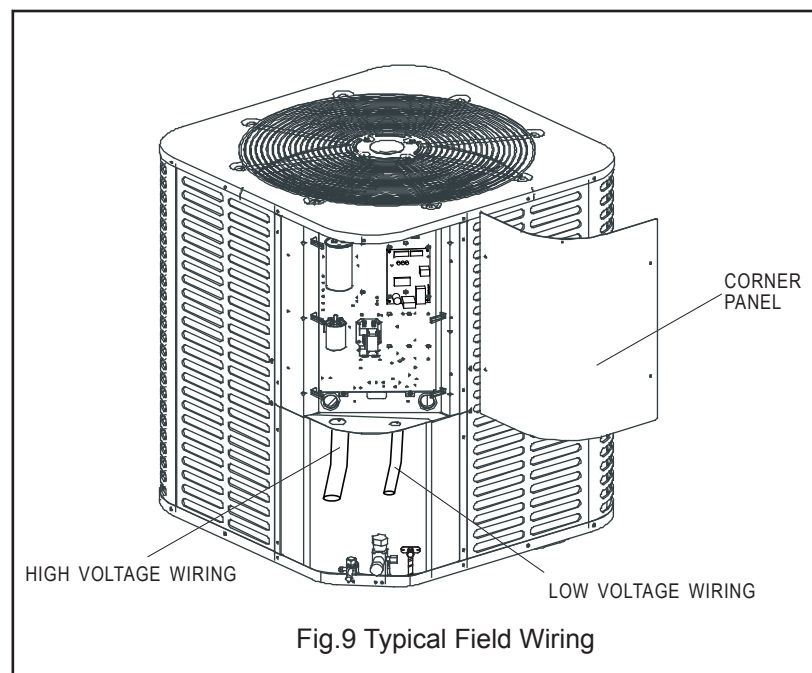
All field wiring must USE COPPER CONDUCTORS ONLY and be in accordance with Local, National Fire, Safety & Electrical Codes. This unit must be grounded with a separate ground wire in accordance with the above codes.

The complete connection diagram and schematic wiring label is located on the inside surface of the unit service access panel and this instruction.

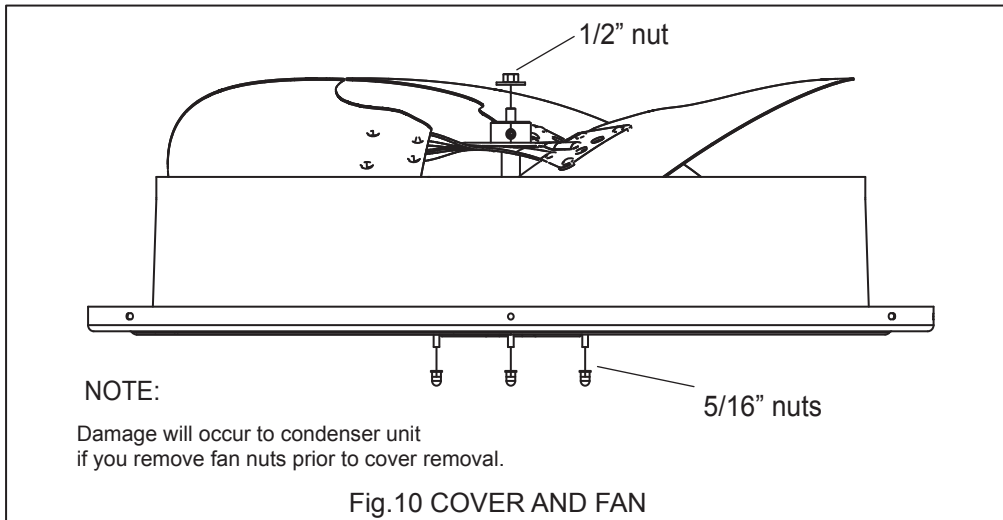
6.2 FIELD CONNECTIONS POWER WIRING

1. Install the proper size weatherproof disconnect switch outdoors and within sight of the unit.
2. Remove the screws at the side of the corner panel. Slide corner panel down and remove from unit. See Fig. 9.
3. Run power wiring from the disconnect switch to the unit.
4. Route wires from disconnect through power wiring opening provided and into the unit control box.
5. Install the proper size time-delay fuses or circuit breaker, and make the power supply connections.
6. Energize the crankcase heater if equipped to save time by preheating the compressor oil while the remaining installation is completed.

NOTE: When changing the motor, remove top cover first.



6.3 REMOVING THE TOP PANEL AND MOTOR



When motor requires changing follow the steps below:

Step 1: Go into electrical panel, disconnect motor power lines.

IMPORTANT NOTE

Disconnect main power to unit. Severe burns and electrical shock will occur if you do not disconnect main power.

Step 2: Remove cover (be careful of motor wires)

Step 3: Be sure to place fan cover unit on the ground as indicated in Fig. 10

IMPOTANT NOTE

Do not place or lean fan blades on ground or against surface.

Step 4: Remove fan motor by removing 5/16" nuts from cover.

Step 5: Remove fan blade from motor by removing 1/2" nut and place fan on the ground.

Step 6: Reverse removal process to reinstall the fan and motor.

IMPOTANT NOTE

When connecting motor wires be sure to check motor direction.

7.0 CHECKING REFRIGERANT CHARGE

Charge for all systems should be checked against the Charging Chart inside the access corner panel or Charging by weight.

IMPORTANT: Do not operate the compressor without charge in system. Addition of R-410A will raise pressures (suction, liquid and discharge).

7.1 CHARGING BY LIQUID PRESSURE

In order to properly charge the system, the following conditions must be met:

- 1) Outdoor temperature above 60°F.
- 2) Indoor temperature between 70°F to 100°F.
- 3) Installation must be complete with brazed joints and drier visually inspected.
- 4) The unit electrical installation must be checked and unit powered for one (1) hour if crank case heater is used or five (5) minutes if no crankcase heater is used.

Follow these steps:

1. Run in cooling mode at least 20 minutes.
2. Measure OUTDOOR AMBIENT TEMPERATURE within 6 inches of coil.
3. Measure SUCTION LINE PRESSURE.
4. Find the TARGET LIQUID PRESSURE at the intersection between the SUCTION LINE PRESSURE and the OUTDOOR AMBIENT TEMPERATURE, if falls between rows or columns then estimate the TARGET LIQUID PRESSURE or SUCTION LINE PRESSURE falls between rows or columns then estimate the TARGET LIQUID PRESSURE between the rows and columns.

5. Compare the measured LIQUID LINE PRESSURE to the TARGET LIQUID PRESSURE, add charge to raise the pressure or recover charge to lower it.
6. After running unit for 20 minutes if the SUCTION LINE PRESSURE changes, go back to step 2 otherwise remove test equipment and cover the valves.

7.2 CHARGING BY WEIGHT

For a new installation, evacuation of interconnecting tubing and indoor coil is adequate; otherwise, evacuate the entire system. The factory refrigerant charge in the outdoor unit is sufficient for 15 feet of standard size interconnecting liquid line. Calculate actual charge required with installed liquid line size and length, please see 4.1 of instruction.

With an accurate scale (+/- 1 oz.) adjust charge difference between that shown on the unit data plate and that calculated for the new system Installation. if the entire system has been evacuated, add the total calculated charge.

7.3 FINAL LEAK TESTING

After the unit has been properly evacuated and charged, a halogen leak detector should be used to detect leaks in the system. All piping within the condensing unit, evaporator, and interconnecting tubing should be checked for leaks. If a leak is detected, the refrigerant should be recovered before repairing the leak. The Clean Air Act prohibits releasing refrigerant into the atmosphere.

8.0 WARRANTY

Assist owner with processing Warranty cards and/or online registration.

8.1 MAINTENANCE

1. Dirt should not be allowed to accumulate on the indoor or outdoor coils or other parts in the air circuit. Clean as often as necessary to keep the unit clean. Use a brush, vacuum cleaner attachment, or other suitable means.
2. The outdoor fan motor is permanently lubricated and does not require periodic oiling.
3. Refer to the furnace or air handler instructions for filter and blower motor maintenance.
4. The indoor coil and drain pan should be inspected and cleaned regularly to assure proper drainage.



CAUTION

It is unlawful to knowingly vent, release or discharge refrigerant into the open air during repair, service, maintenance or the final disposal of this unit. When the system is functioning properly and the owner has been fully instructed, secure the owner's approval.

9.0 WIRING DIAGRAM



CAUTION

These units must be wired and installed in accordance with all National and Local Safety Codes.

9.1 CONTROL WIRING

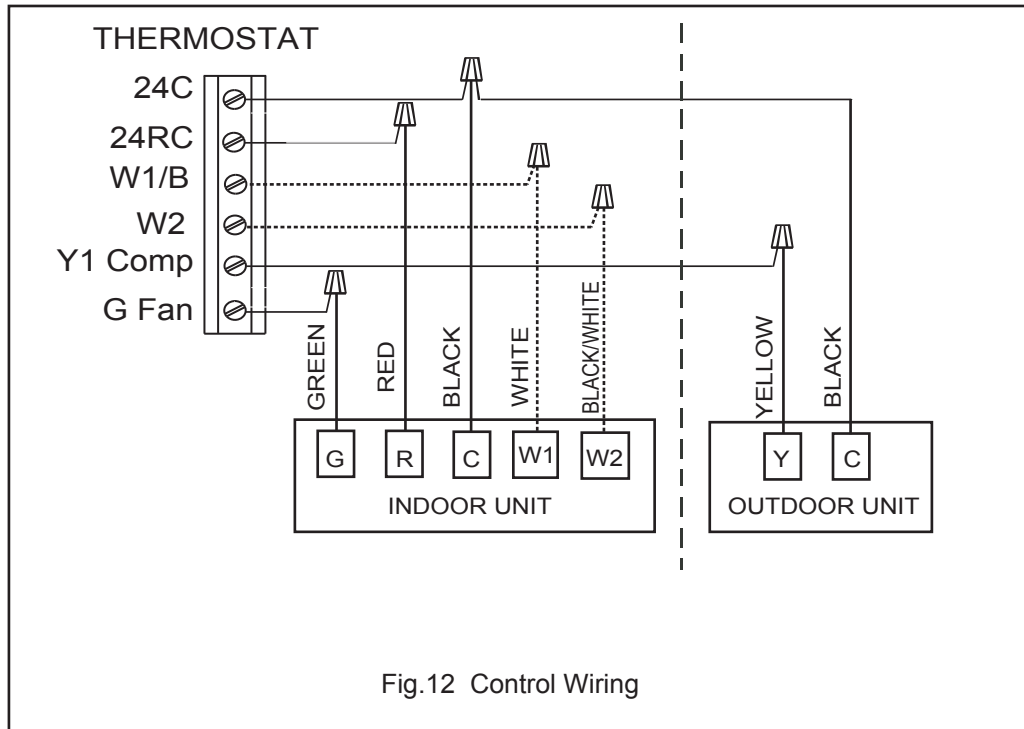
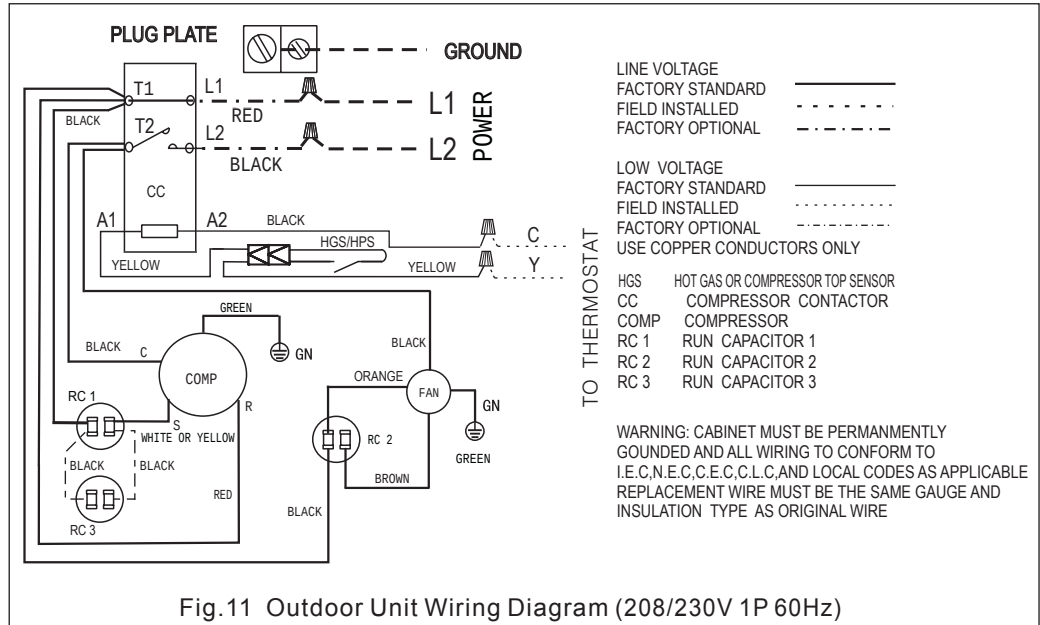


TABLE 4: Electrical Data:

Model	Minimum Circuit Ampacity(A)	Maximum Circuit Protector(A)
18	10.4	15
24	14.4	25
30	15.2	25
36	19.9	35
42	28.2	45
43	21.6	35
48	32.0	50
60	31.1	50

Orifice/TXV Replacement Information

NOTE: The orifice / TXV replacement options noted in this sheet supersede those in the installation guide. Please reference this sheet for all refrigerant metering options.

Step 1: Remove the screws and front coil panel.

Step 2: Remove the rubber plugs from the liquid and vapor lines.

Step 3: Using a back up wrench and loosen the flow assembly. Remove the factory installed piston Reassemble and tighten the flow assembly. Tighten the flow assembly to 11 (±2) ft-lb

Step 4: If replacing the preinstalled orifice with the optional orifice, ensure the replacement orifice is placed in the same orientation as the preinstalled orifice before tightening the flow assembly.

TXV-Specific Steps

Step 5: Drill a small hole in the suction line for the TXV equalizer line. The hole must be on top as noted in the picture.

Step 6: Dry fit the TXV to the liquid line entering the coil. The valve must be in the upright position as pictured. Keep the valve as close to the coil as possible, although a short piece of field fabricated tubing may be needed. The TXV must be mounted in the CORRECT direction of flow. Place the TXV equalizer line 3 to 4 mm inside the small hole drilled in the vapor line.

Step 7: Wrap the TXV and coil panel with a wet rag to prevent overheating while brazing. Use a nitrogen flow and braze all connections.

Step 8: Allow tubing to cool and pressurize line sets with 150 PSI of nitrogen to check braze connections and flow assembly for leaks. Make repairs as needed.

Step 9: Locate and clean a straight section of the vapor line as close to the coil as possible. Use the supplied copper straps to secure the TXV sensing bulb on top of the vapor line as pictured.

Step 10: Insulate the entire vapor line and sensing bulb. It is also recommended to insulate the TXV and liquid line between the valve and coil to prevent condensation in hot humid environments.

Step 11: Replace the front coil panel and secure in place.

Step 12: Follow the steps in the installation guide for vacuum requirements and system start up procedures.

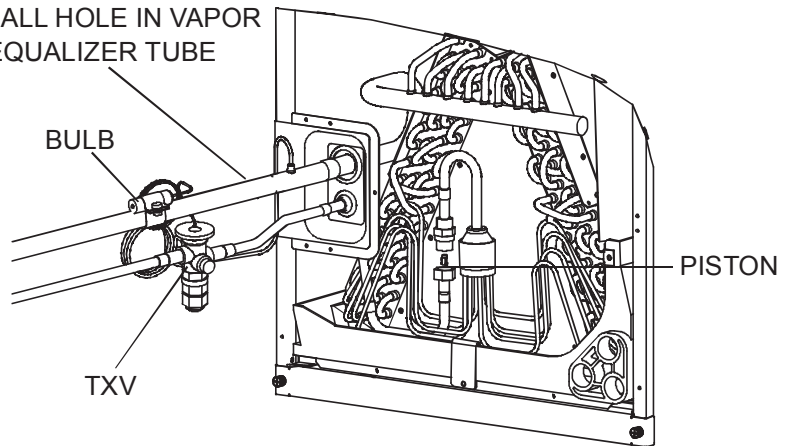
Step 13: Allow system to run for a minimum of 10 minutes in the cooling mode.

A: For fixed orifice installation: Use the superheat charging chart to obtain proper superheat based on indoor / outdoor conditions.

B: For TXV installation: Charge the system to 10 degrees of subcooling. Adjust the expansion valve to achieve 9 (± 3 degrees) of superheat.

Step 14: Allow system to run for an additional 10 minutes to verify the subcooling and superheat readings.

DRILL A SMALL HOLE IN VAPOR LINE FOR EQUALIZER TUBE



Superheat Charging Chart

Outdoor Temp (°F)	Indoor Temperature (°F) Dry Bulb/Wet Bulb					
	95/79	90/75	85/71	80/67	75/63	70/58
115	23	16	7	5	5	5
110	24	17	9	5	5	5
105	26	19	11	5	5	5
100	27	21	13	7	5	5
95	29	23	16	9	5	5
90	30	25	18	12	5	5
85	32	26	20	14	8	5
80	34	28	22	17	11	5
75	35	30	24	19	13	6
70	37	32	26	21	16	10
65	38	34	29	24	19	13
60	40	36	31	27	22	17
55	41	37	33	29	25	21

Note 1: Chart is based on 400 CFM/Ton indoor airflow and 50% relative humidity. If indoor relative humidity is above 70% or below 20%, use indoor wet bulb temperature only. Airflow range is 375 to 425 CFM/Ton.

System Charge Adjustment

Subcooling Charging (Above 55°F Outdoor Temp.)

R-410A SUBCOOLING CHARGING TABLE		
OUTDOOR TEMPERATURE (°F)		
82	95	115
DESIGN SUBCOOLING VALUES (°F)		
10~12	8~10	5~8

Condensing Unit		Air Handle / A-Coil	Piston Size	Optional TXV	SEER	If Preinstalled Piston Replace Required
AC	M4AC3018C1000NA	M4CXC024BA1CAAA	C050	MAYTXVACAC1818AA	13	YES
		M4CXC025BB1CAAA	C050	MAYTXVACAC1818AA	13	YES
		M4AH3018B1000AA	C050	MAYTXVACAC1818AA	13	YES
	M4AC3024B1000NA	M4CXC024BA1CAAA	C056	MAYTXVACHP1830AA	13	NO
		M4CXC025BB1CAAA	C056	MAYTXVACHP1830AA	13	NO
		M4AH3024B1000AA	C056	MAYTXVACHP1830AA	13	NO
	M4AC3030B1000NA	M4CXC024BA1CAAA	C060	MAYTXVACHP1830AA	13	YES
		M4CXC025BB1CAAA	C060	MAYTXVACHP1830AA	13	YES
		M4AH3030B1000AA	C063	MAYTXVACHP1830AA	13	YES
	M4AC3036B1000NA	M4CXC035BA1CAAA	C069	MAYTXVACHP3642AA	13	NO
		M4CXC035BB1CAAA	C067	MAYTXVACHP3642AA	13	NO
		M4CXC035BC1CAAA	C067	MAYTXVACHP3642AA	13	NO
		M4AH3036B1000AA	C071	MAYTXVACHP3642AA	13	YES
	M4AC3042B1000NA	M4CXC042BB1CAAA	C074	MAYTXVACHP3642AA	13	NO
		M4CXC048BC1CAAA	C074	MAYTXVACHP3642AA	13	YES
		M4CXC060BD1CAAA	C074	MAYTXVACHP3642AA	13	YES
		M4AH3042B1000AA	C072	MAYTXVACHP3642AA	13	YES
	M4AC3043B1000NA	M4CXC042BB1CAA	C075	MAYTXVACHP3642AA	13.5	YES
		M4CXC048BC1CAA	C075	MAYTXVACHP3642AA	13.5	YES
		M4CXC060BD1CAA	C075	MAYTXVACHP3642AA	13.5	YES
		M4AH3042B1000AA	C075	MAYTXVACHP3642AA	13.5	NO
	M4AC3048B1000NA	M4CXC042BB1CAAA	C078	MAYTXVACHP4848AA	13	YES
		M4CXC048BC1CAAA	C078	MAYTXVACHP4848AA	13	NO
		M4CXC060BD1CAAA	C078	MAYTXVACHP4848AA	13	NO
		M4AH3048B1000AA	C078	MAYTXVACHP4848AA	13	YES
	M4AC3060B1000NA	M4CXC049BC1CAAA	C088	MAYTXVACHP6060AA	13	NO
		M4CXC061BD1CAAA	C088	MAYTXVACHP6060AA	13	NO
		M4AH3060B1000AA	C090	MAYTXVACHP6060AA	13	YES

*The optional TXV means the recommended model for field installation.

REFRIGERANT CHARGE FOR AC SYSTEM

Cooling Mode		13 SEER R410A AC Charge Chart 1.5 TON													
		Outdoor Ambient Temperature(°F)													
		55	60	65	70	75	80	85	90	95	100	105	110	115	
Suction Pressure at Large Service Valve(psig)		Liquid Pressure at Small Service Valve(psig)													
	165			276	294	314	335	354	376	396	421	443	467	496	
	161			274	292	312	333	352	374	394	419	441	465	494	
	157			272	290	310	331	350	372	392	417	439	463	492	
	153		246	270	288	308	329	348	370	390	415	437	461	490	
	149		244	268	286	306	327	346	368	388	413	435	459	488	
	145		242	266	284	304	325	344	366	386	411	433	457	486	
	141	230	240	264	282	302	323	342	364	384	409	431	455	484	
	137	228	238	262	280	300	321	340	362	382	407	429	453	482	
	133	226	236	260	278	298	319	338	360	380	405	427	451	480	
	129	224	234	258	276	296	317	336	358	378	403	425	449	478	
	125	222	232	256	274	294	315	334	356	376	401	423	447	476	
	121	220	230	254	272	292	313	332	354	374	399	421	445	474	
	117	218	228	252	270	290	311	330	352	372	397	419	443	472	
	113	216	226	250	268	288	309	328	350	370	395	417	441	470	
	109	214	224	248	266	286	307	326	348	368	393	415	439	468	
105	212	222	246	264	284	305	324	346	366	391	413	437	466		

Cooling Mode		13 SEER R410A AC Charge Chart 2.0 TON												
		Outdoor Ambient Temperature(°F)												
		55	60	65	70	75	80	85	90	95	100	105	110	115
		Liquid Pressure at Small Service Valve(psig)												
Suction Pressure at Large Service Valve(psig)	165			269	290	310	331	353	375	397	422	446	471	495
	161			267	288	308	329	351	373	395	420	444	469	493
	157			265	286	306	327	349	371	393	418	442	467	491
	153		242	263	284	304	325	347	369	391	416	440	465	489
	149		240	261	282	302	323	345	367	389	414	438	463	487
	145		238	259	280	300	321	343	365	387	412	436	461	485
	141	216	236	257	278	298	319	341	363	385	410	434	459	483
	137	214	234	255	276	296	317	339	361	383	408	432	457	481
	133	212	232	253	274	294	315	337	359	381	406	430	455	479
	129	210	230	251	272	292	313	335	357	379	404	428	453	477
	125	208	228	249	270	290	311	333	355	377	402	426	451	475
	121	206	226	247	268	288	309	331	353	375	400	424	449	473
	117	204	224	245	266	286	307	329	351	373	398	422	447	471
	113	202	222	243	264	284	305	327	349	371	396	420	445	469
	109	200	220	241	262	282	303	325	347	369	394	418	443	467
	105	198	218	239	260	280	301	323	345	367	392	416	441	465

Cooling Mode		13 SEER R410A AC Charge Chart 2.5 TON													
		Outdoor Ambient Temperature(°F)													
		55	60	65	70	75	80	85	90	95	100	105	110	115	
Suction Pressure at Large Service Valve(psig)	Liquid Pressure at Small Service Valve(psig)														
	165			269	290	312	333	355	378	400	426	452	477	503	
	161			267	288	310	331	353	376	398	424	450	475	501	
	157			265	286	308	329	351	374	396	422	448	473	499	
	153		242	263	284	306	327	349	372	394	420	446	471	497	
	149		240	261	282	304	325	347	370	392	418	444	469	495	
	145		238	259	280	302	323	345	368	390	416	442	467	493	
	141	214	236	257	278	300	321	343	366	388	414	440	465	491	
	137	212	234	255	276	298	319	341	364	386	412	438	463	489	
	133	210	232	253	274	296	317	339	362	384	410	436	461	487	
	129	208	230	251	272	294	315	337	360	382	408	434	459	485	
	125	206	228	249	270	292	313	335	358	380	406	432	457	483	
	121	204	226	247	268	290	311	333	356	378	404	430	455	481	
	117	202	224	245	266	288	309	331	354	376	402	428	453	479	
	113	200	222	243	264	286	307	329	352	374	400	426	451	477	
	109	198	220	241	262	284	305	327	350	372	398	424	449	475	
105	196	218	239	260	282	303	325	348	370	396	422	447	473		

Cooling Mode		13 SEER R410A AC Charge Chart 3.0 TON													
		Outdoor Ambient Temperature (°F)													
		55	60	65	70	75	80	85	90	95	100	105	110	115	
Suction Pressure at Large Service Valve(psig)	Liquid Pressure at Small Service Valve(psig)														
	165			277	298	318	339	363	386	410	436	461	487	512	
	161			275	296	316	337	361	384	408	434	459	485	510	
	157			273	294	314	335	359	382	406	432	457	483	508	
	153		250	271	292	312	333	357	380	404	430	455	481	506	
	149		248	269	290	310	331	355	378	402	428	453	479	504	
	145		246	267	288	308	329	353	376	400	426	451	477	502	
	141	224	244	265	286	306	327	351	374	398	424	449	475	500	
	137	222	242	263	284	304	325	349	372	396	422	447	473	498	
	133	220	240	261	282	302	323	347	370	394	420	445	471	496	
	129	218	238	259	280	300	321	345	368	392	418	443	469	494	
	125	216	236	257	278	298	319	343	366	390	416	441	467	492	
	121	214	234	255	276	296	317	341	364	388	414	439	465	490	
	117	212	232	253	274	294	315	339	362	386	412	437	463	488	
	113	210	230	251	272	292	313	337	360	384	410	435	461	486	
	109	208	228	249	270	290	311	335	358	382	408	433	459	484	
105	206	226	247	268	288	309	333	356	380	406	431	457	482		

Cooling Mode		13(42) SEER R410A AC Charge Chart 3.5 TON												
		Outdoor Ambient Temperature(°F)												
		55	60	65	70	75	80	85	90	95	100	105	110	115
Suction Pressure at Large Service Valve(psig)		Liquid Pressure at Small Service Valve(psig)												
	165			265	286	306	327	349	372	394	421	449	476	503
	161			263	284	304	325	347	370	392	419	447	474	501
	157			261	282	302	323	345	368	390	417	445	472	499
	153		238	259	280	300	321	343	366	388	415	443	470	497
	149		236	257	278	298	319	341	364	386	413	441	468	495
	145		234	255	276	296	317	339	362	384	411	439	466	493
	141	212	232	253	274	294	315	337	360	382	409	437	464	491
	137	210	230	251	272	292	313	335	358	380	407	435	462	489
	133	208	228	249	270	290	311	333	356	378	405	433	460	487
	129	206	226	247	268	288	309	331	354	376	403	431	458	485
	125	204	224	245	266	286	307	329	352	374	401	429	456	483
	121	202	222	243	264	284	305	327	350	372	399	427	454	481
	117	200	220	241	262	282	303	325	348	370	397	425	452	479
	113	198	218	239	260	280	301	323	346	368	395	423	450	477
	109	196	216	237	258	278	299	321	344	366	393	421	448	475
	105	194	214	235	256	276	297	319	342	364	391	419	446	473

Cooling Mode		13(43) SEER R410A AC Charge Chart 3.5 TON													
		Outdoor Ambient Temperature(°F)													
		55	60	65	70	75	80	85	90	95	100	105	110	115	
		Liquid Pressure at Small Service Valve(psig)													
Suction Pressure at Large Service Valve(psig)	165			250	269	288	307	329	351	373	400	426	453	479	
	161			248	267	286	305	327	349	371	398	424	451	477	
	157			246	265	284	303	325	347	369	396	422	449	475	
	153		227	244	263	282	301	323	345	367	394	420	447	473	
	149		225	242	261	280	299	321	343	365	392	418	445	471	
	145		223	240	259	278	297	319	341	363	390	416	443	469	
	141	204	221	238	257	276	295	317	339	361	388	414	441	467	
	137	202	219	236	255	274	293	315	337	359	386	412	439	465	
	133	200	217	234	253	272	291	313	335	357	384	410	437	463	
	129	198	215	232	251	270	289	311	333	355	382	408	435	461	
	125	196	213	230	249	268	287	309	331	353	380	406	433	459	
	121	194	211	228	247	266	285	307	329	351	378	404	431	457	
	117	192	209	226	245	264	283	305	327	349	376	402	429	455	
	113	190	207	224	243	262	281	303	325	347	374	400	427	453	
	109	188	205	222	241	260	279	301	323	345	372	398	425	451	
	105	186	203	220	239	258	277	299	321	343	370	396	423	449	

Cooling Mode		13 SEER R410A AC Charge Chart 4 TON												
		Outdoor Ambient Temperature(°F)												
		55	60	65	70	75	80	85	90	95	100	105	110	115
Suction Pressure at Large Service Valve(psig)		Liquid Pressure at Small Service Valve(psig)												
	165			257	277	298	318	340	362	384	411	439	466	493
	161			255	275	296	316	338	360	382	409	437	464	491
	157			253	273	294	314	336	358	380	407	435	462	489
	153		231	251	271	292	312	334	356	378	405	433	460	487
	149		229	249	269	290	310	332	354	376	403	431	458	485
	145		227	247	267	288	308	330	352	374	401	429	456	483
	141	204	225	245	265	286	306	328	350	372	399	427	454	481
	137	202	223	243	263	284	304	326	348	370	397	425	452	479
	133	200	221	241	261	282	302	324	346	368	395	423	450	477
	129	198	219	239	259	280	300	322	344	366	393	421	448	475
	125	196	217	237	257	278	298	320	342	364	391	419	446	473
	121	194	215	235	255	276	296	318	340	362	389	417	444	471
	117	192	213	233	253	274	294	316	338	360	387	415	442	469
	113	190	211	231	251	272	292	314	336	358	385	413	440	467
	109	188	209	229	249	270	290	312	334	356	383	411	438	465
	105	186	207	227	247	268	288	310	332	354	381	409	436	463

Cooling Mode		13 SEER R410A AC Charge Chart 5 TON												
		Outdoor Ambient Temperature(°F)												
		55	60	65	70	75	80	85	90	95	100	105	110	115
Suction Pressure at Large Service Valve(psig)		Liquid Pressure at Small Service Valve(psig)												
	165			265	285	306	326	348	369	391	416	441	466	491
	161			263	283	304	324	346	367	389	414	439	464	489
	157			261	281	302	322	344	365	387	412	437	462	487
	153		239	259	279	300	320	342	363	385	410	435	460	485
	149		237	257	277	298	318	340	361	383	408	433	458	483
	145		235	255	275	296	316	338	359	381	406	431	456	481
	141	212	233	253	273	294	314	336	357	379	404	429	454	479
	137	210	231	251	271	292	312	334	355	377	402	427	452	477
	133	208	229	249	269	290	310	332	353	375	400	425	450	475
	129	206	227	247	267	288	308	330	351	373	398	423	448	473
	125	204	225	245	265	286	306	328	349	371	396	421	446	471
	121	202	223	243	263	284	304	326	347	369	394	419	444	469
	117	200	221	241	261	282	302	324	345	367	392	417	442	467
	113	198	219	239	259	280	300	322	343	365	390	415	440	465
	109	196	217	237	257	278	298	320	341	363	388	413	438	463
	105	194	215	235	255	276	296	318	339	361	386	411	436	461

TABLE 5. Troubleshooting

SYSTEM FAULTS	WHAT TO CHECK MODE																									
	HIGH VOLTAGE WIRING	COMPRESSOR CAPACITOR	I.D. FAN CAPACITOR	CONTRACTOR CAPACITOR	LOW VOLTAGE WIRING	CONTROL TRANSFORMER	LOW VOLTAGE FUSE	STUCK COMPRESSOR	REF. UNDERCHARGE	EXCESSIVE EVAP. LOAD	NONCONDENSABLES	RES. O.D. AIRFLOW	O.D. AIR RECIRCULATION	TXV STUCK OPEN	REF. CIR. RESTRICTIONS	RES. I.D. AIRFLOW	SUPERRIANT	SOV. COIL DEFECTIVE	CHECK VALVE LEAKING	DEFROST CONTROL DEF.	T4 TEMP. SENSOR DEF.	T3 TEMP. SENSOR DEF.	HPC/HS. SENSOR DEF.	HI TEMP. SENSOR DEF.		
REFRIGERANT CIRCUIT																										
Head Pressure Too High	C														P	P	S	P	S							
	H														P	P										
Head Pressure Too Low	C									S	P					S	S		P	S	S		P			
	H									S	P					S	S		S	S	S	P				
Suction Pressure Too High	C									S		P	P				S			P		P				
	H									S							S			P						
Suction Pressure Too Low	C										P						S	P	S							
	H										P			S	S		S		S		S					
Liquid Refrig. Floodback (TXV)	C															P					P					
	H															P					P					
I.D. Coil Frosting	C										P			S	S											
	H																									
Compressor Runs Inadequate or No Cooling/Heating	C									S	P		S	S			S	P	S	S		S				
	H									S	P		S				S	P	S	S	S	S				
ELECTRICAL																										
Compressor & O.D. Fan Won't Start	C	P	P				S	S	P	S	P	P														
	H	P	P				S	P	S		P											S	S	S	S	S
Compressor Will Not Start But O.D. Fan Runs	C	P	P									P														
	H	P	P				S				P	P										S		S		
O.D. Fan Won't Start	C	P			P																					
	H	P			P																	S				
Compressor Hums But Won't Start	C			P			S					P														
	H			P			S					P														
I.D. Blower Won't Start	C	P	P	S			P		S	P	S		S													
	H	P	P	S			P		S	P	S		S													
DEFROST																										
Unit Won't Initiate Defrost	C																									
	H																				P		P		S	
Defrost Terminates on Time	C																									
	H										P												P		S	
Unit Icing Up	C																									
	H										P			S	S		S		P		P					

C- Cooling H - Heating P - Primary Causes S - Secondary Causes

TABLE 6. Operational And Checkout Procedures

Final phases of this installation are the unit Operational and Checkout Procedures. To obtain proper performance, all units must be operated and charge adjustments made in accordance with procedures found in the Service Facts of the Outdoor Unit.

After installation has been completed, it is recommended that the entire system be checked against the following list:

1. Be sure unit suspension(if used) is secure and that there are no tools or loose debris in or around or on top of the unit..... []
2. Properly insulate suction lines and fittings. []
3. Properly secure and isolate all refrigerant lines... []
4. Verify that all electrical connections are tight. []
5. Check all duct outlets; they must be open and unrestricted. []
6. Check drain lines and be sure all joints are tight..... []
7. Be sure that a return air filter is installed. []
8. Operate complete system in each mode to verify proper performance. Verify operation of supplementary electric heater. []

Base Limited Warranty

Single Phase R-410A Outdoor Units, Single Phase R-22 Outdoor Units, Air Handlers, Furnaces, Packaged Units, Cased Coils, Ductless and Specific Terms for Commercial Applications.

Subject to the terms and conditions of this limited warranty, Ingersoll Rand ("Company") extends a limited warranty against manufacturing defects for the product(s) identified in **Table 1** attached hereto ("Products") that are installed in a residential/multi-family application (personal, family or household purposes) under normal use and maintenance in the United States and Canada.

In order to maximize the available benefits under this limited warranty, the Purchaser (as defined below) should read it in its entirety. All repairs of Product parts covered under this limited warranty must be made with authorized service parts and by a licensed HVAC service provider. Additionally, commercial applications are treated differently under this limited warranty as stated in **Table 1** attached hereto. For purposes of this limited warranty, "commercial applications" shall mean any application other than for personal, family, or household use.

TERM: The limited warranty period for Products is as stated in **Table 1** attached hereto. If the Purchaser properly registers the Products, the limited warranty period shall be extended as stated in **Table 1** attached hereto. Regardless of registration, the Commencement Date for a limited warranty period shall be the date that the original installation is complete and all Product start-up procedures have been properly completed and verified by an installer's invoice. If the installation and start-up date cannot be verified by the installer's invoice, the Commencement Date shall be sixty (60) days after the factory manufacture date which is verified by the Product serial number. Where a Product is installed in a newly constructed home, the Commencement Date is the date the Purchaser purchased the residence from the builder. Proof of Product purchase, installation, and/or closing date of the residence may be required to confirm the Commencement Date.

The installation of Product replacement parts under this limited warranty shall not extend the original warranty period. The warranty period for any Product part replaced under this limited warranty is the applicable warranty period remaining under the original Product warranty.

WHO IS COVERED: This limited warranty is provided only to the original owner and his or her spouse ("Purchaser") of the residence where the Products are originally installed. This warranty is not transferable except according to terms stated on the applicable website identified below under Registration Requirements. Company has the right to request any and all proof of Product purchase or installation and/or closing date of the residence.

WHAT COMPANY WILL DO: Company may request proof of Product purchase and/or installation in order to provide Product parts under this limited warranty. As Company's only responsibility and Purchaser's only remedy under this limited warranty, Company will furnish a replacement part to the licensed HVAC service provider, without charge for the part only, to replace any Product part that fails due to a manufacturing defect under normal use and maintenance. The Purchaser must pay for any and all shipping and handling charges and other costs of warranty service for the replacement part. If a Product part is not available, Company will, at its option, provide a free suitable substitute part or provide a credit in the amount of the then factory selling price for a new suitable substitute part to be used by the Purchaser towards the retail purchase price of a new Company product. Any new Product purchase shall be at Purchaser's sole cost and expense including, but not limited to, all shipping, removal, and installation costs and expenses.

REGISTRATION REQUIREMENTS: All Products must be properly registered online by the Purchaser within sixty (60) days after the Commencement Date to receive the registered limited warranty terms. To register online, go to:

<http://www.ameristarhvac.com>

and click "Begin Online Registration." If a Purchaser does not register within this stated time period, the base limited warranty terms shall apply.

ELIGIBILITY REQUIREMENTS: The following items are required in order for the Products to be covered under this limited warranty:

- The Products must be in the same location where they were originally installed.
- The Products must be properly installed, operated, and maintained by a licensed HVAC service provider in accordance with the Product specifications or installation, operation, and maintenance instructions provided by Company with each Product. Failure to conform to such specifications and/or instructions shall void this limited warranty. Company may request written documentation showing the proper preventative maintenance.
- All Product parts replaced by Company under this limited warranty must be given to the servicing provider for return to Company.
- Air handlers, air conditioners, heat pumps, cased or uncased coils, stand-alone furnaces, and packaged units must be part of an Air Conditioning, Heating, and Refrigeration Institute rated and matched system or a specification in a Company provided bulletin or otherwise approved in writing by a Company authorized representative.

EXCLUSIONS: The following are not covered by this limited warranty:

- Labor costs including, but not limited to, costs for diagnostic calls or the removal and reinstallation of Products and/or Product parts.
- Shipping and freight expenses required to ship Product replacement parts.
- Failures, defects, or damage (including, but not limited to, any loss of data or property) caused by (1) any third party product, service, or system connected or used in conjunction with the Products; (2) any use that is not designed or intended for the Products; (3) modification, alteration, abuse, misuse, negligence, or accident; (4) improper storage, installation, maintenance, or operation including, but not limited to, operation of electrical equipment at voltages other than the range specified on the Product nameplate; (5) any use in violation of written instructions or specifications provided by Company; (6) any acts of God including, but not limited to, fire, water, storms, lightning, or earthquakes; or any theft or riots; or (7) a corrosive atmosphere or contact with corrosive materials such as, but not limited to, chlorine, fluorine, salt, sulfur, recycled waste water, urine, fertilizers, rust, or other damaging substances or chemicals.
- Products purchased direct including, but not limited to, Internet or auction purchases and purchases made on an uninstalled basis.
- Cabinets or cabinet pieces that do not affect product performance, air filters, refrigerant, refrigerant line sets, belts, wiring, fuses, surge protection devices, non-factory installed driers, and Product accessories.
- Increased utility usage costs.

REFRIGERANT POLICY: (1) Manufacturer-Installed Refrigerant: Beginning on January 1, 2010, R-22 refrigerant will no longer be used as a manufacturer-installed refrigerant as required by federal regulation. All Products with manufacturer-installed refrigerant will include R410-A refrigerant. Any and all expenses or costs associated with replacing Product parts that are not R-410A compatible will not be covered by the terms and conditions of this limited warranty. (2) Non-Manufacturer installed Refrigerant: For Products manufactured and sold by the Company without refrigerant, only manufacturer approved and genuine alternate refrigerants shall be used. The use of contaminated, counterfeit, non-genuine, or non-manufacturer approved alternate refrigerant will void this limited warranty. (3) All Products: Products include a liquid line filter drier which must be replaced when a compressor replacement is necessary. A suction line filter drier must be added for compressors defined as burnouts and failure to do so will void this warranty. Non-approved refrigerant and/or non-approved refrigerant system additives including, but not limited to dyes will void this limited warranty.

ADDITIONAL TERMS:

THIS LIMITED WARRANTY AND LIABILITY SET FORTH HEREIN ARE IN LIEU OF ALL OTHER WARRANTIES AND LIABILITIES, WHETHER IN CONTRACT OR IN NEGLIGENCE, EXPRESS OR IMPLIED, IN LAW OR IN FACT. THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE LIMITED TO THE DURATION OF THE APPLICABLE PRODUCT WARRANTY. COMPANY DOES NOT AUTHORIZE ANY PERSON TO CREATE FOR IT ANY OBLIGATION OR LIABILITY IN CONNECTION WITH THE PRODUCTS.

NOTWITHSTANDING ANYTHING IN THIS LIMITED WARRANTY TO THE CONTRARY, COMPANY SHALL NOT BE LIABLE FOR ANY INCIDENTAL, CONSEQUENTIAL, INDIRECT, SPECIAL AND/OR PUNITIVE DAMAGES, WHETHER BASED ON CONTRACT, WARRANTY, TORT (INCLUDING, BUT NOT LIMITED TO, STRICT LIABILITY OR NEGLIGENCE), PATENT INFRINGEMENT, OR OTHERWISE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. COMPANY'S MAXIMUM LIABILITY HEREUNDER IS LIMITED TO THE ORIGINAL PURCHASE PRICE OF THE PRODUCTS.

No action arising out of any claimed breach of this limited warranty may be brought by a Purchaser more than one (1) year after the cause of action has arisen.

This limited warranty gives you specific legal rights, and you may also have other rights as otherwise permitted by law. If this Product is considered a consumer product, please be advised that some local laws do not allow limitations on incidental or consequential damages, how long a warranty lasts based on registration, or how long an implied warranty lasts, so that the above limitations may not fully apply. Refer to your local laws for your specific rights under this limited warranty. If you have any questions regarding this limited warranty, please contact your original installation dealer, or any participating dealer, should your original installation dealer no longer be available.

TABLE 1: Warranty Time Periods

COVERAGE TERMS FOR RESIDENTIAL APPLICATIONS:

Pursuant to the Ingersoll Rand ("Company") limited warranty terms and conditions, the following Products are covered for the base time periods as stated below ("Base Limited Warranty Period"). If registered, the Base Limited Warranty Periods for certain Products will be extended as stated below ("Registered Limited Warranty Period").

FURNACES:

M801P, M951P, M952V

Base Limited Warranty Period: Parts – five (5) years, Heat Exchanger – twenty (20) years.

Registered Limited Warranty Period: Parts – ten (10) years, Heat Exchanger: twenty (20) years.

A801X, A951X, A952V

Base Limited Warranty Period: Parts – five (5) years, Heat Exchanger – twenty (20) years.

AIR HANDLERS:

M4AH3, M4AH4, M4AH6

Base Limited Warranty Period: Indoor Coil and Parts – five (5) years.

Registered Limited Warranty Period: Indoor Coil and Parts – ten (10) years.

TMM4, TMM5

Base Limited Warranty Period: Indoor Coil and Parts – three (3) years.

Registered Limited Warranty Period: Indoor Coil and Parts – ten (10) years.

ELECTRIC HEATERS, installed in M4A3/4 AIR HANDLERS:

MAYHTR1A

Limited Warranty Period: Parts – ten (10) years.

SINGLE PHASE R410 OUTDOOR UNITS:

M4AC3 (**M4AC3018, 24, 30, 36, 42, 43, 48, 60 only**), M4AC6, M4HP3, M4AC4, M4HP4, M4HP6

Base Limited Warranty Period: Compressor, Outdoor Coil, Parts: five (5) years.

Registered Limited Warranty Period: Compressor, Outdoor Coil, Parts – ten (10) years.

SINGLE PHASE R410 OUTDOOR UNITS:

M4AC3 (**M4AC3017, 23, 29 only**)

Base Limited Warranty Period: Compressor, Outdoor Coil, Parts: one (1) year.

Registered Limited Warranty Period: Compressor, Outdoor Coil, Parts – five (5) years.

SINGLE PHASE R22 OUTDOOR UNITS:

M2AC3, M2HP3

Base Limited Warranty Period: Compressor, Outdoor Coil, Parts - one (1) year.

Registered Limited Warranty Period: Compressor, Outdoor Coil, Parts – five (5) years.

CASED COILS:

M4CXC

Base Limited Warranty Period: Coil, Parts –five (5) years.

Registered Limited Warranty Period: Coil and Parts – ten (10) years.

4MXC, 4GXC

Base Limited Warranty Period: Coil, Parts –three (3) years.

Registered Limited Warranty Period: Coil and Parts – ten (10) years.

PACKAGED UNITS:

M4PH3, M4PG3, M4PH4, M4PW4, M4PG4

Base Limited Warranty Period: Compressor, Coil, Parts – five (5) years, Heat Exchanger – ten (10) years.

Registered Limited Warranty Period: Compressor, Coil, Parts – ten (10) years, Heat Exchanger – ten (10) years.

4TCA4, 4WCA4

Base Limited Warranty Period: Compressor, Coil, Parts – five (5) years

Registered Limited Warranty Period: Compressor, Coil, Parts – seven (7) years

ELECTRIC HEATERS, installed in M4PH4 PACKAGED UNITS:

MAYHTR1P

Limited Warranty Period: Parts – ten (10) years.

ELECTRIC HEATERS, installed in 4TCA4/4WCA4 PACKAGED UNITS:

BAYHTR1V

Limited Warranty Period: Parts – five (5) years.

DUCTLESS SYSTEMS:**

M4THS/M4MHW, M4TCS/M4MCW, M4MLW/M4TLS, M4THM:

Base Limited Warranty Period:

All Parts – five (5) year.

Registered Limited Warranty Period:

All Parts – ten (10) years.

**There is no distinction between residential and commercial use for this Limited Warranty term and coverage.

SPECIFIC TERMS FOR COMMERCIAL APPLICATIONS

Base Limited Warranty Period Applies

此页不做菲林，只作说明

V1.0升V1.1

更改型号AA为NA, P23

V1.1升V1.2

更新保修卡内容

V1.2升V1.3

更新保修卡内容，更改14位编码

V1.3升1.4

更新P5：18改为12，P6:18改成24

V1.4升1.5

更改封面认证，更改保修卡

V1.5升V1.6

更改保修卡

V1.6升V1.7

更新保修卡

删除说明书冷暖信息，调整版面

V1.7升V1.8

更新保修卡

P5/P11/P17增加43参数

P19/20/21增加43阀芯、冷媒充注表，版面更改

V1.8升V1.9

更新保修卡

V1.9升V2.0

P17：修改图11

V2.0升V. J

更新保修卡（2019.4.28）

V. J升V. K

去掉FIG. 2中过线孔尺寸标注

双胶纸100g

大A4规格

印刷颜色为黑白